ModelIt!

Student Activity Pack

Why Do Some Neighborhoods Flood While Others Don't?

Modeling Urban Water Systems and Environmental Justice

Grade 11L1 | NGSS HS-ESS3-4, HS-ETS1-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever experienced flooding in your neighborhood or seen it on the news? What do you think determines which areas flood and which stay dry?**

2. **Why do you think some neighborhoods have more parks and trees while others are mostly concrete and asphalt?**

3. **What happens to rainwater when it falls on a parking lot versus a forest? Draw a diagram showing where the water goes in each case.**

4. **Do you think all neighborhoods in a city have the same quality of infrastructure (roads, pipes, drainage)? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Impervious Surface Coverage | Storm Drain Capacity | Green Infrastructure Area | Runoff Volume | Flood Depth

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Disinvested Neighborhood

Impervious: 90% | Storm drains: Low capacity (1950s) | Green infrastructure: Near zero

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Affluent Neighborhood

Impervious: 60% | Storm drains: High capacity (modern) | Green infrastructure: Moderate

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Green Infrastructure Retrofit

Disinvested neighborhood + significant green infrastructure addition

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Historical Redlining Score — A measure of how severely a neighborhood was disinvested due to discriminatory housing and lending policies from the 1930s-1970s — neighborhoods graded 'D' (hazardous) by the HOLC received less infrastructure investment for decades

Climate Change Intensification — The increasing frequency and intensity of extreme rainfall events due to climate change — a warming atmosphere holds 7% more moisture per degree Celsius, making the storms that overwhelm aging infrastructure more common

Community Displacement Risk — The likelihood that flood damage will force residents to permanently relocate — lower-income renters without flood insurance face the highest displacement risk, fundamentally changing neighborhood demographics after major floods

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When a storm hits a city, why do low-income neighborhoods and communities of color flood first — and is the infrastructure designed that way on purpose? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that flood risk is a result of infrastructure decisions rather than natural conditions?**

2. **What does the historical connection between redlining and current flood vulnerability reveal about how past racism creates present environmental injustice?**

3. **How could your model be used by city planners to prioritize flood mitigation investments in the most equitable way?**

4. **Why is it important to include community residents in flood mitigation planning rather than designing solutions for them without their input?**

5. **What limitations does your model have in capturing the full human impact of flooding beyond property damage?**

8 STEM Challenge: Design an Equitable Urban Flood Mitigation Plan

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YOUR ENGINEERING MISSION

Design a flood mitigation plan for a city with historically unequal infrastructure that achieves equitable flood protection across all neighborhoods regardless of income or demographics.

THE SCENARIO

A city of 500,000 people has experienced three catastrophic floods in five years, each disproportionately damaging the same low-income neighborhoods and communities of color. The city council has allocated $200 million for flood mitigation and hired your environmental engineering team to design an equitable plan. Historical analysis shows that the most-flooded neighborhoods have 90% impervious surfaces, 1950s-era storm drains, and almost no green space, while affluent neighborhoods have modern infrastructure and abundant tree canopy.

Engineering Design Thinking

1. WHAT SPECIFIC GREEN INFRASTRUCTURE FEATURES WILL YOU INSTALL IN THE HIGHEST-RISK NEIGHBORHOODS, AND HOW MUCH RUNOFF WILL EACH ABSORB?

2. HOW WILL YOU BALANCE STORM DRAIN REPLACEMENT WITH GREEN INFRASTRUCTURE — WHICH PROVIDES MORE FLOOD REDUCTION PER DOLLAR?

3. WHAT ROLE DOES REDUCING IMPERVIOUS SURFACE COVERAGE PLAY IN YOUR PLAN, AND IS IT FEASIBLE IN DENSE URBAN NEIGHBORHOODS?

4. HOW WILL YOU ENGAGE THE RESIDENTS OF AFFECTED NEIGHBORHOODS IN THE DESIGN PROCESS TO ENSURE THE PLAN MEETS THEIR NEEDS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Engineers design stormwater management systems, green infrastructure, and flood mitigation projects, earning $70,000–$130,000/year. Environmental Justice Analysts work at nonprofits, government agencies, and consulting firms to ensure infrastructure investments address historical inequities, earning $55,000–$110,000/year. Urban Planners who specialize in climate resilience earn $60,000–$120,000/year designing cities that protect all communities.